Target Brazil

E-Commerce Operations Analysis

3-Year Strategic Diagnostic · 2016 – 2018 · 100,000+ Orders

99,441	R$16.0M	R$161	27
Total Orders	Total Revenue	Avg Order Value	States Covered

Prepared by	Dataset	Region	Tools
Sadi Jagdish Reddy Data Analyst	8 Datasets · 100K+ Orders	Brazil · 27 States · 4,119 Cities	Python · SQL · Excel · Tableau

Executive Summary

3-Year performance overview and key strategic findings

This report analyzes **99,441 orders** placed on Target Brazil between September 2016 and October 2018 — a 25-month window capturing the early growth phase of one of the world's largest retailers entering the Brazilian e-commerce market. The dataset spans 8 interconnected tables covering customers, orders, order items, payments, products, sellers, reviews, and geolocation across all **27 Brazilian states** and **4,119 cities**.

The headline finding is a **+137% revenue growth** comparing January-August 2017 to the same period in 2018 — extraordinary expansion that reflects both genuine market growth and improving platform maturity. However, this growth is not evenly distributed. **São Paulo alone accounts for 40%+ of all orders**, creating a geographic concentration that represents both a strength and a strategic vulnerability. Meanwhile, remote northern states face freight costs 3-4x higher than São Paulo — a barrier that is actively suppressing demand in underpenetrated markets.

Customer satisfaction is strong — an average review score of **4.09/5.0** — and delivery performance frequently exceeds estimates. Credit card dominates at 74%+ of payments, with installment plans widely used — a behavioral pattern specific to Brazilian consumer culture that has significant implications for product pricing and checkout optimization strategy.

Total Orders	Total Revenue	Avg Order Value	Avg Delivery
99,441	R$16.0M	R$161	12.1 days

Avg Review Score	Unique Cities	Total Sellers	Revenue Growth
4.09 / 5.0	4,119	3,095	+137%

Order Volume & Revenue Growth

3-year trend analysis across the full dataset period

Annual Order Volume — A Clear Growth Trajectory

Figure 1.1 — Annual Order Volume 2016–2018

The annual order data tells an unambiguous story: Target Brazil was growing rapidly across the full three-year window. 2016 data covers only the final quarter of the year, making direct year-on-year comparison imperfect — but the trajectory from 2017 to 2018 is clear. Order volume grew significantly, and the pace of growth was accelerating rather than plateauing. This is characteristic of a marketplace that has reached product-market fit and is now scaling — the hard early work of building seller relationships, customer trust, and logistics infrastructure is paying compounding dividends.

Monthly Seasonality — Structural Patterns Across Three Years

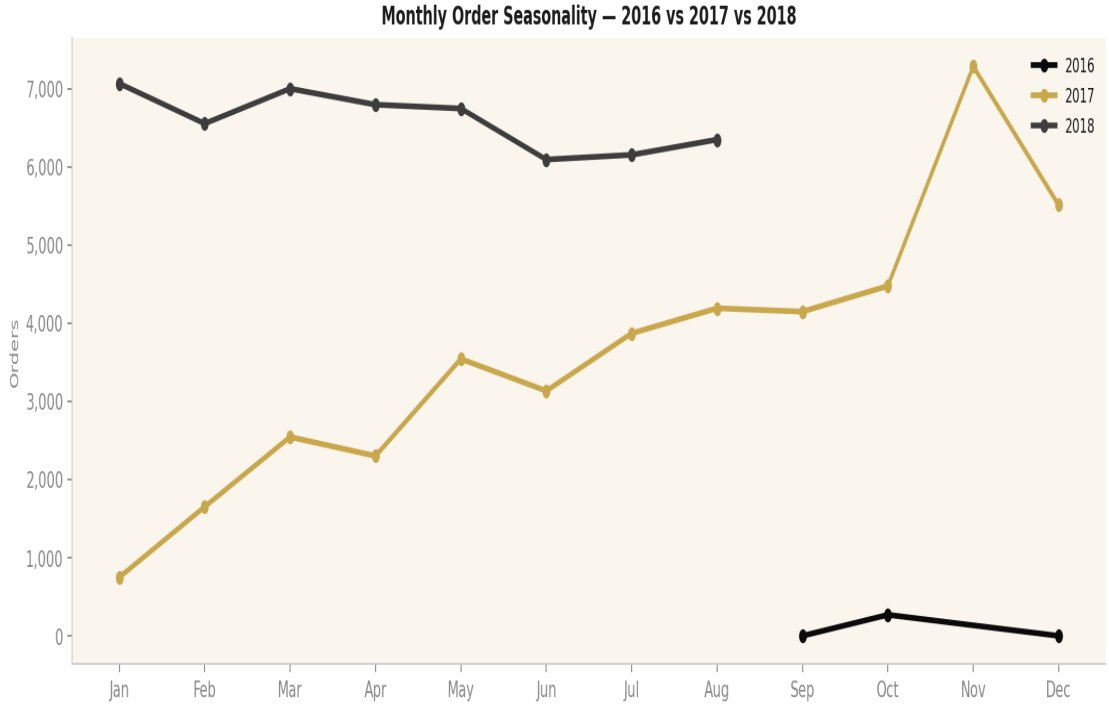

Figure 1.2 — Monthly Order Seasonality 2016–2018

Plotting monthly orders across all three years reveals seasonality patterns that are structural rather than coincidental. November spikes dramatically in both 2017 and 2018 — the Black Friday effect is clear and consistent. December holds strong. January dips in the post-holiday period. What is most useful for operations planning is not the peak itself but the ramp — orders begin accelerating in October, which means inventory, logistics capacity, and seller preparation should be in place by mid-September at the latest. The brands and sellers who miss this window lose the season.

Customer Geography & Market Distribution

State-level analysis across all 27 Brazilian states

Customer Distribution by State

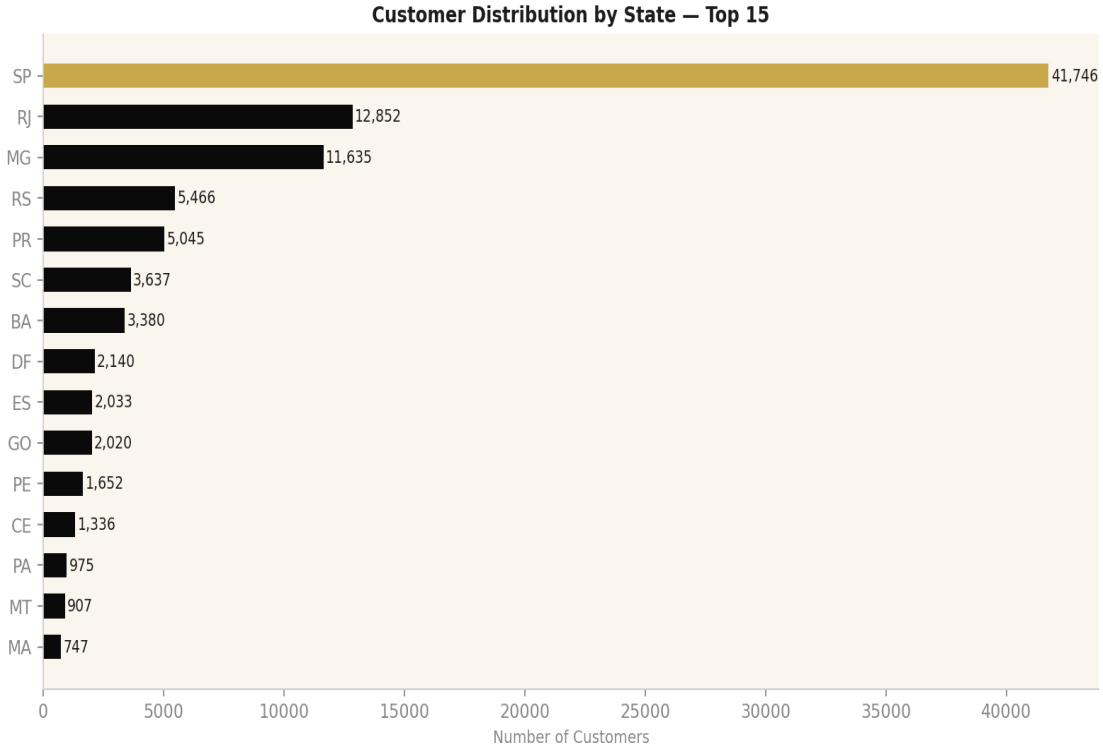

Figure 2.1 — Customer Distribution by State (Top 15)

São Paulo dominates the customer map — not just as the largest state but by a margin that reflects the extraordinary concentration of Brazil's consumer economy in the southeast. SP, RJ, and MG together account for approximately 60% of all customers. This concentration is both an asset and a risk. It means Target Brazil can generate enormous volume from a geographically compact area with well-developed logistics. But it also means the business is highly exposed to anything that disrupts consumer confidence or purchasing power in São Paulo specifically.

Month-on-Month Orders by State — Top 5 Markets

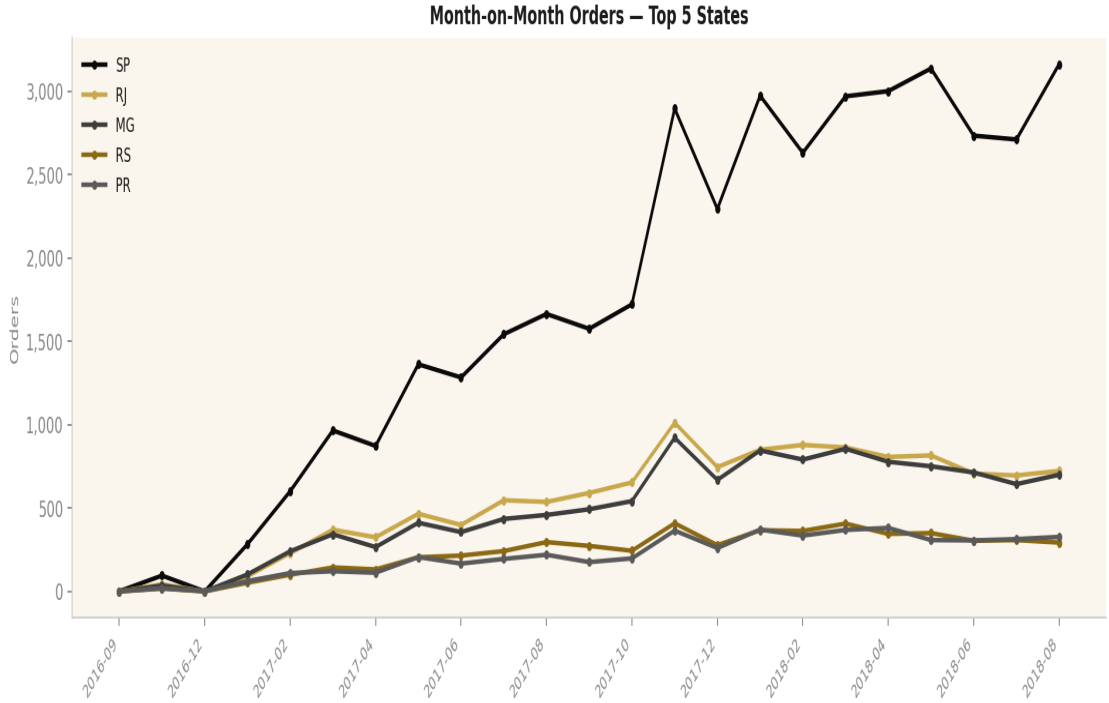

Figure 2.2 — Month-on-Month Orders — Top 5 States

Tracking the top 5 states month-by-month reveals that São Paulo's growth is genuinely exceptional — it pulls away from the other four states throughout the analysis period. Rio de Janeiro and Minas Gerais maintain a consistent second and third tier. What is strategically interesting is whether this gap is closing or widening — because if SP is growing faster than all other states combined, the concentration risk is increasing rather than naturally resolving through platform growth.

Economic Impact & Revenue Analysis

Payment values, order pricing and freight economics

Revenue Growth 2017 → 2018 — The 137% Story

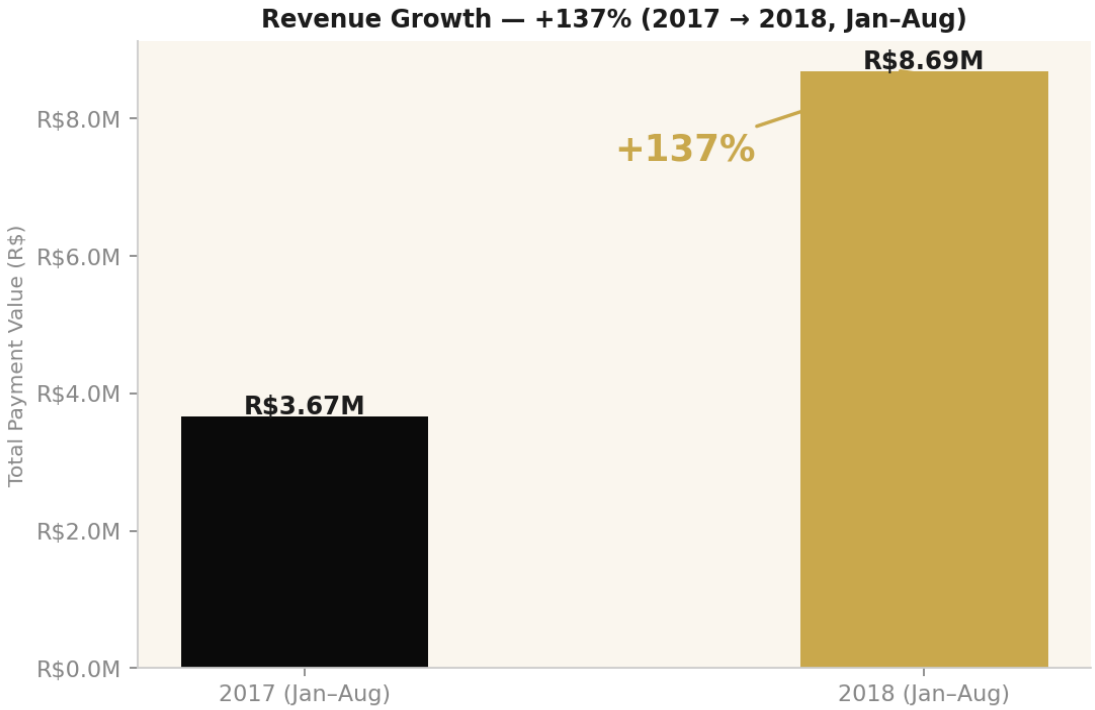

Figure 3.1 — Revenue Growth Jan–Aug 2017 vs 2018

Comparing January to August 2017 against the same period in 2018 produces a number that demands attention: **+137% revenue growth**. This is not organic consumer behaviour compounding quietly — this is platform inflection. In the space of twelve months, Target Brazil more than doubled its revenue run rate. The drivers are likely multi-factorial: more sellers onboarded, more products available, improved search and discovery, growing brand awareness, and an expanding customer base benefiting from word-of-mouth. Understanding which of these factors drove the most growth would require seller-level and traffic data, but the headline outcome is unambiguous.

Order Price and Freight Analysis by State

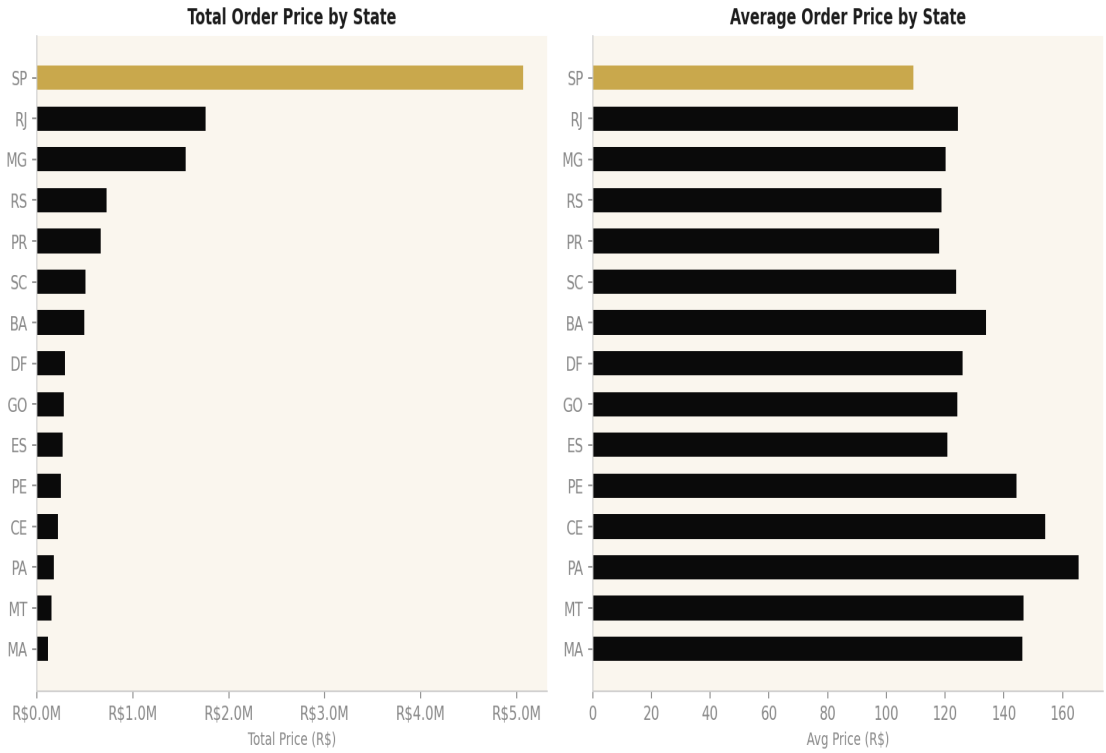

Figure 3.2 — Total and Average Order Price by State

São Paulo generates the highest total order revenue by a significant margin — driven by volume rather than average order value. When you look at average price per order, the picture becomes more nuanced. States like DF (Distrito Federal — Brasília) and RJ show higher average order values, suggesting wealthier customer demographics or a stronger preference for premium product categories. This has implications for category expansion: premium electronics, luxury goods, and high-end fashion may over-index in DF relative to its total order volume.

Freight Economics — The Hidden Barrier to National Growth

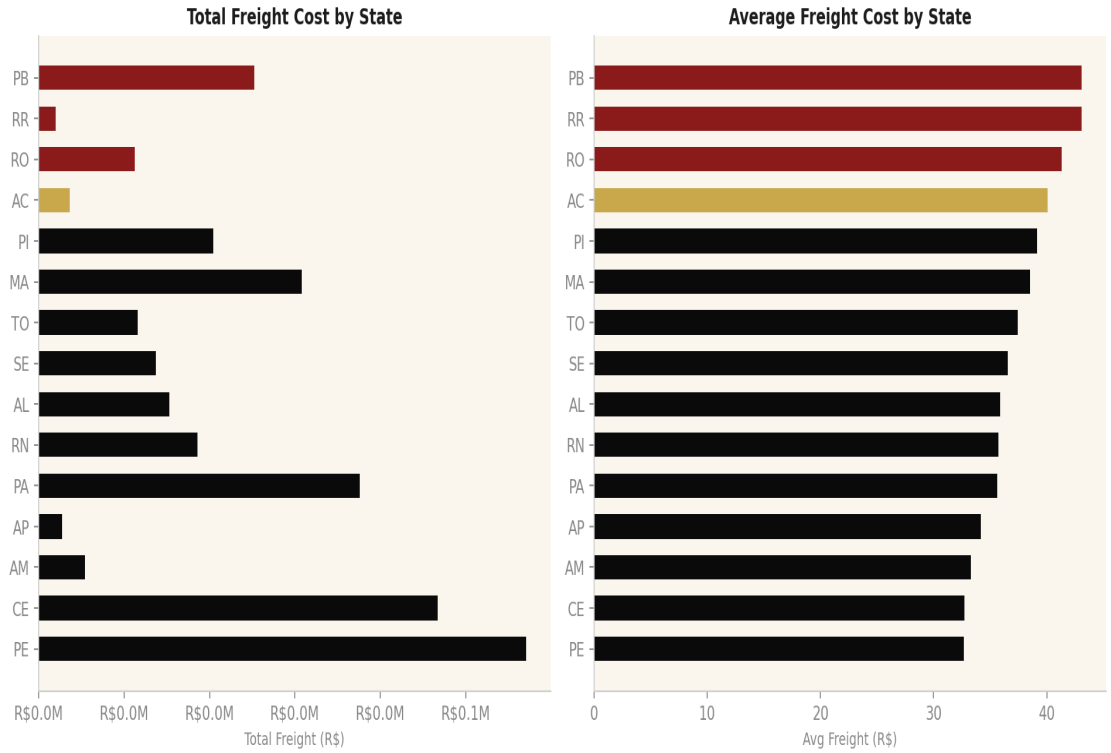

Figure 3.3 — Total and Average Freight Cost by State

The freight data exposes one of the most significant structural barriers to Target Brazil's national expansion. Remote northern states — Roraima, Amapá, Amazonas — face average freight costs 3 to 4 times higher than São Paulo. For a R$150 product, a R$40 freight charge is a 27% surcharge that materially changes the purchase decision. Many customers in these states will see the freight cost, compare it to the product price, and abandon the cart. This is not a customer preference problem — it is a logistics economics problem that requires a structural solution: regional carrier partnerships, micro-fulfillment centers in high-freight regions, or a freight subsidy program funded from platform marketing budgets rather than customer invoices.

Delivery Performance Analysis

Speed, accuracy and geographic variation across 27 states

Delivery Time — Fastest vs Slowest States

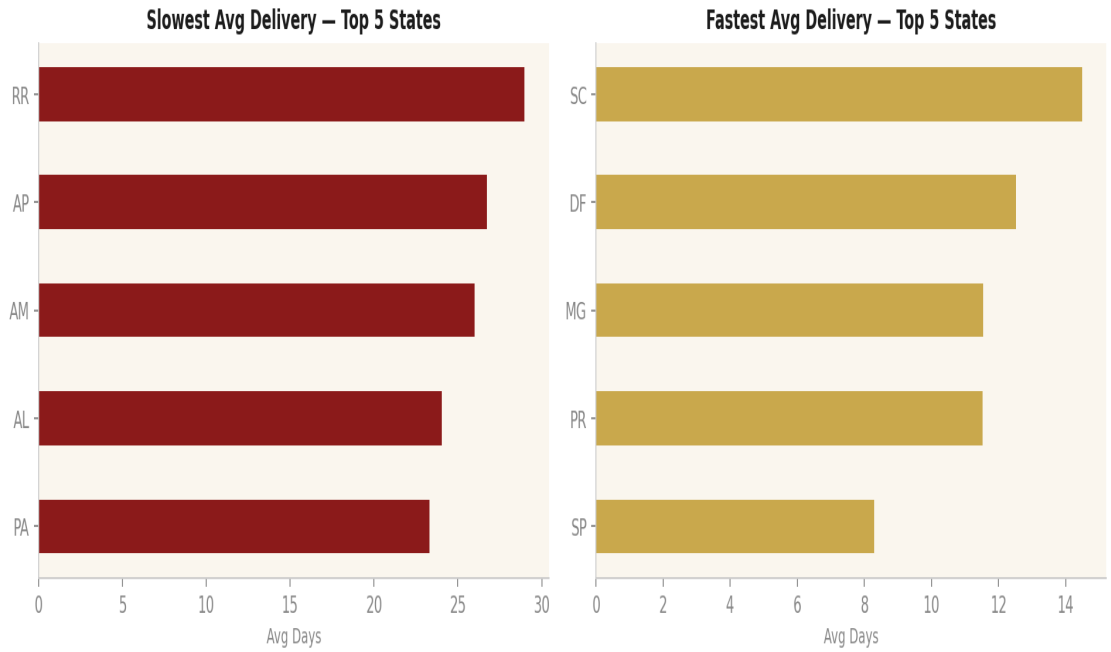

Figure 4.1 — Slowest and Fastest Avg Delivery by State

Delivery time variance across Brazil's 27 states is enormous — ranging from under 8 days in the fastest states to over 25 days in the slowest. This variance is not random. It tracks closely with geographic distance from São Paulo, where the majority of sellers and fulfillment infrastructure is concentrated. Northern and northeastern states consistently appear at the slow end, while southeastern and southern states cluster at the fast end. The operational implication is clear: achieving national delivery consistency requires regional fulfillment — not just faster carriers on existing routes.

States Delivering Fastest vs Estimated Date

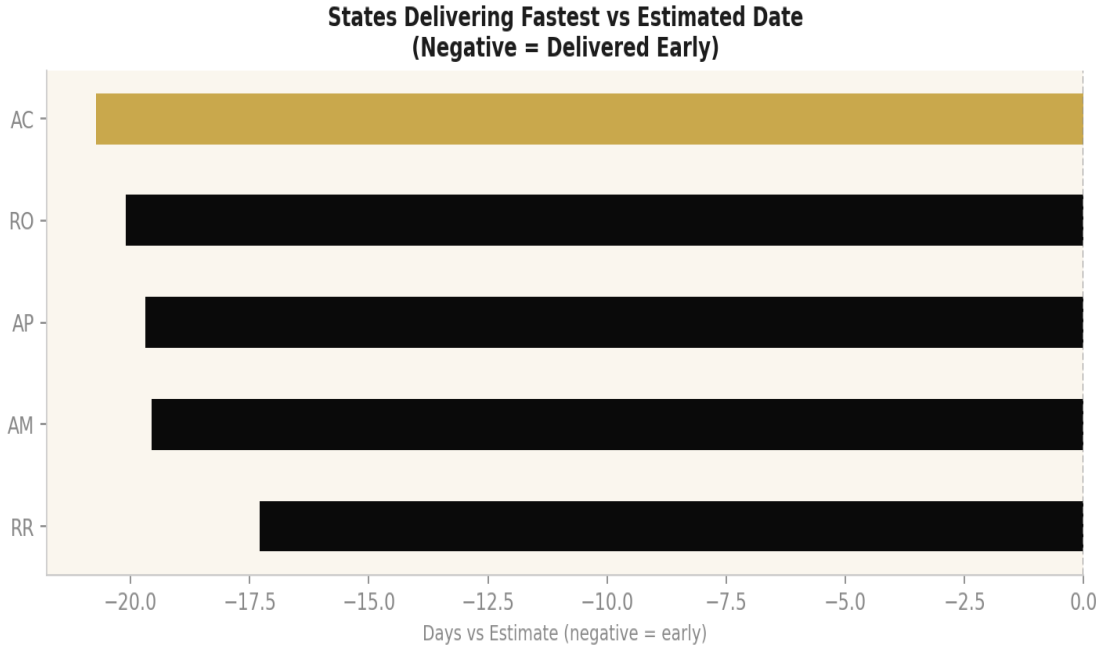

Figure 4.2 — States Consistently Delivering Ahead of Estimate

Several states consistently deliver orders significantly before the estimated delivery date — in some cases 15 to 20 days ahead of the estimate shown to the customer. This is both impressive logistics performance and a missed marketing opportunity. When a customer expects their order in 20 days and it arrives in 5, that surprise creates positive emotional impact that directly drives repeat purchases and reviews. Currently, this extraordinary performance is invisible — the customer sees the estimated date, not the actual. Publishing more accurate (faster) estimates in these states, or explicitly marketing early delivery performance, would convert logistics excellence into commercial advantage.

Freight Top and Bottom 5 States

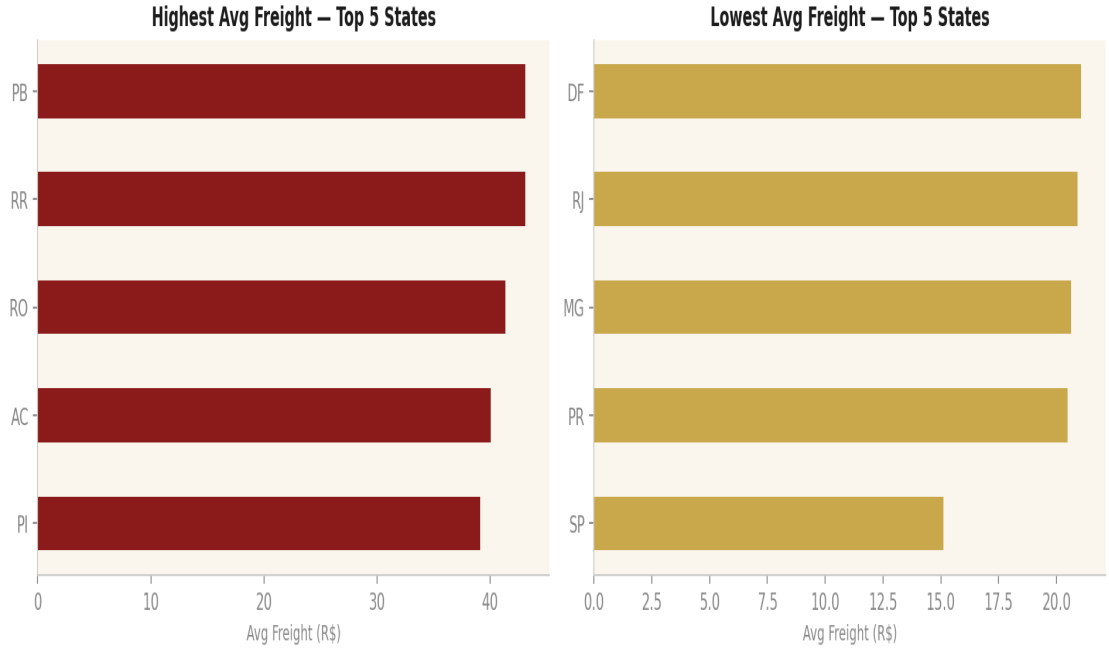

Figure 4.3 — Highest and Lowest Avg Freight by State

The freight comparison between the highest and lowest cost states quantifies the inequality in the Brazilian logistics landscape. The highest freight states are predominantly in the north — remote, less served by national carriers, and far from major distribution hubs. The lowest freight states are in the dense southeast where competition between carriers is highest and routes are most efficient. Addressing this gap is not just a customer experience issue — it is a market expansion imperative.

Payment Behaviour Analysis

Payment types, installment patterns and month-on-month trends

Payment Type Distribution — Credit Card Dominance

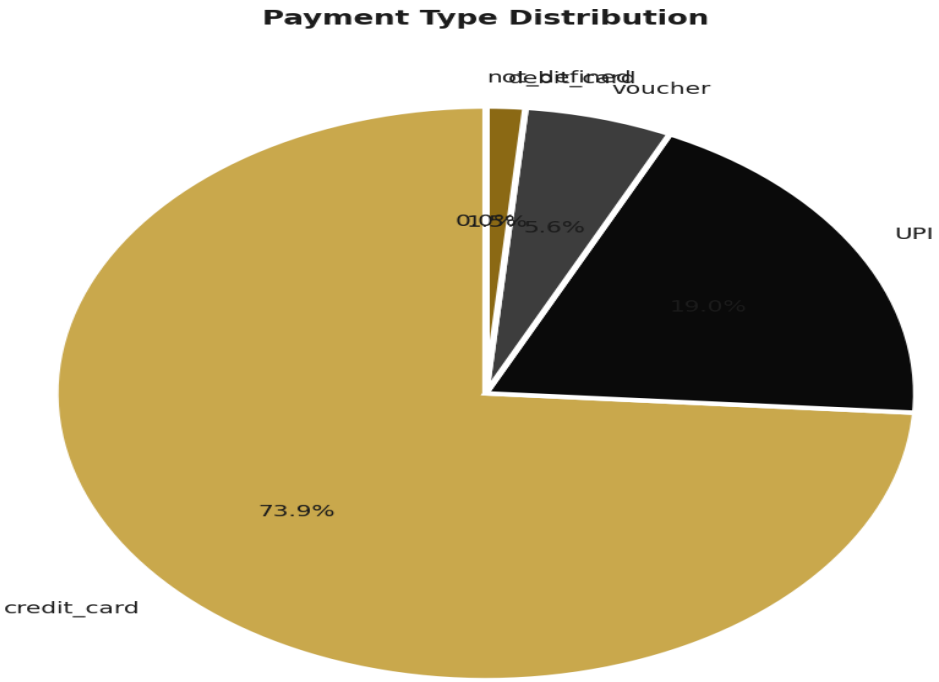

Figure 5.1 — Payment Type Distribution

Credit card accounts for over 74% of all payments — a dominance that reflects both platform design choices and Brazilian consumer culture, where credit card installment plans are deeply embedded in how people budget for larger purchases. Boleto bancário (bank slip) represents a meaningful secondary channel, primarily serving customers without credit cards — a segment that skews toward lower income brackets and is disproportionately concentrated in smaller cities and remote states. Removing or deprioritizing Boleto would directly exclude this segment from the platform.

Payment Installments — The Brazilian Consumer Behavior Pattern

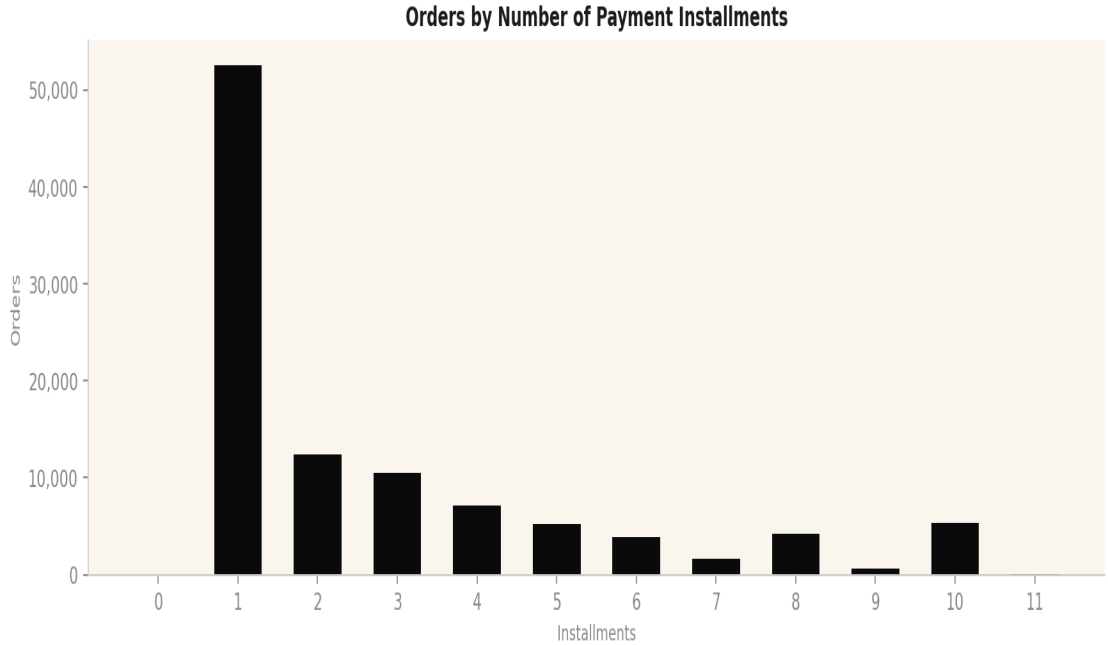

Figure 5.2 — Orders by Number of Installments

The installment data reveals a distinctly Brazilian purchasing pattern. While single-payment orders are most common, a substantial portion of customers split payments into 2, 3, 4, or more installments. This behavior is not driven by inability to pay — it is a cultural preference for cash flow management deeply rooted in how Brazilian retail has operated for decades. The practical implication: offering interest-free installments on high-value products removes one of the most significant psychological barriers to purchase and consistently increases conversion rates on items above R$200.

Month-on-Month Orders by Payment Type

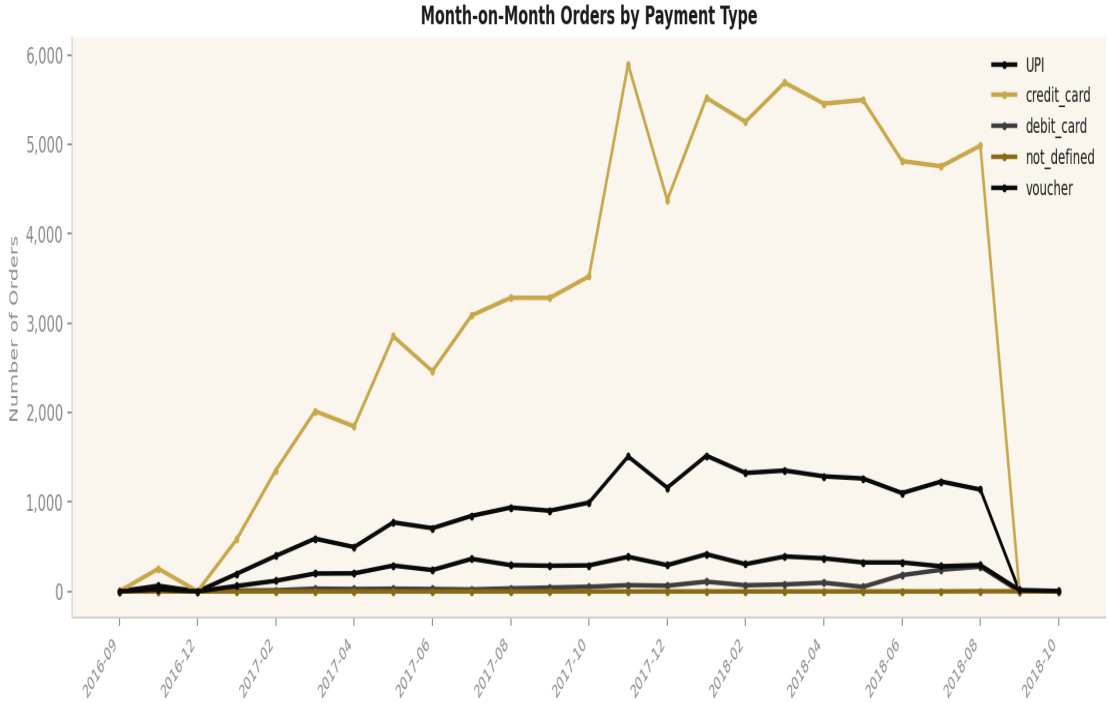

Figure 5.3 — Month-on-Month Orders by Payment Type

Credit card usage grows consistently month-over-month throughout the analysis period, tracking the overall platform growth trajectory almost perfectly. Boleto usage remains relatively stable, suggesting it serves a core customer segment that is not converting to credit card over time. Voucher and debit card usage is minimal throughout. The consistency of these patterns across the full 25-month window suggests that payment preferences at Target Brazil are structural rather than driven by promotional activity or seasonal factors.

Customer Behaviour Patterns

When Brazilians shop and review score distribution

Time of Day — When Brazilian Customers Place Orders

Figure 6.1 — Orders by Time of Day

Brazilian customers predominantly place orders during the **Afternoon (13-18h)**, followed by Night (19-23h). Morning orders are significant but trail both afternoon and night. Dawn (0-6h) represents a small but non-trivial volume — likely driven by customers browsing after midnight when they cannot sleep, or in time zones with different peak hours. The practical implication for Target Brazil's marketing team is direct: promotional push notifications, flash sale launches, and email campaigns should be timed to reach customers at the start of their peak browsing window — around 12:30 to 13:00 — to maximize the chance of converting browse intent into a completed purchase during the afternoon peak.

Customer Review Score Distribution

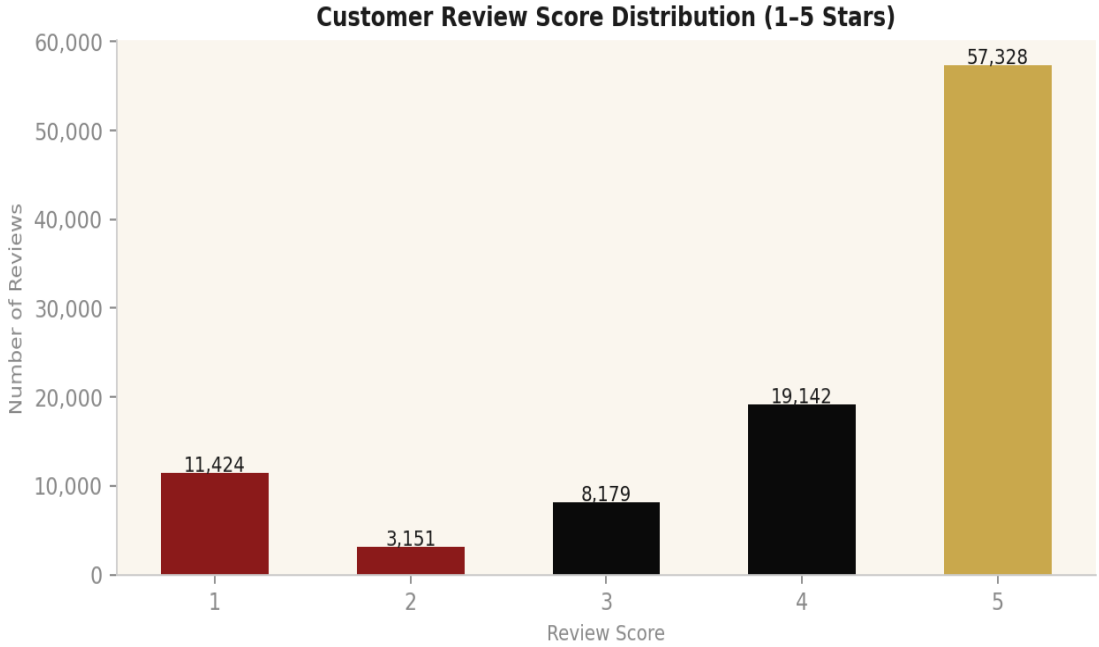

Figure 6.2 — Review Score Distribution (1–5 Stars)

The review distribution is strongly polarized toward the extremes — 5-star reviews are by far the most common, followed by 1-star reviews as the second most frequent score. This bimodal pattern is typical of marketplace review systems: satisfied customers leave 5 stars, significantly disappointed customers leave 1 star, and most customers in between do not review at all. The 4.09 average is healthy but masks a meaningful minority of severely negative experiences — likely concentrated in late deliveries, wrong items, or seller quality issues. Identifying the sellers and product categories that generate disproportionate 1-star reviews and addressing those specifically would have an outsized impact on the overall platform score.

Strategic Recommendations

Eight priorities ranked by revenue and market impact

The following recommendations connect directly to specific data findings in this report. Each is actionable within 30-90 days with existing resources and addresses a quantified problem or opportunity in the Target Brazil operation.

CRITICAL · Build Regional Fulfillment Capacity Before Next Peak Season

The freight disparity between northern and southeastern states is suppressing demand in underpenetrated markets. Freight costs in RR, AP, and AM are 3-4x higher than SP. A regional micro-fulfillment center in Manaus would service the Amazon basin at dramatically reduced freight rates. Alternatively, negotiating regional carrier contracts for the north and northeast — funded partially from marketing budgets as customer acquisition cost — would materially improve conversion rates in markets where Target Brazil currently has low penetration and relatively no competition.

CRITICAL · Launch Inventory and Logistics Planning 8 Weeks Before November

The Black Friday and holiday season spike is reliable and large — 2.5 to 3x normal monthly revenue. Yet the operational window to prepare is consistently underutilized. Sellers need to be briefed on expected volume by mid-September. Carrier capacity needs to be pre-negotiated by October 1. Inventory of top-selling SKUs needs to be built by October 15. A platform that can sustain the November-December peak without stockouts, delivery delays, or seller failures will compound its reputation advantage season over season.

HIGH · Introduce Free Freight Threshold for High-Freight States

The single most impactful customer acquisition tool available for remote states is a free freight threshold — a spend level above which shipping is subsidized by the platform. Model the incremental revenue needed from each high-freight state to break even on a R$20-30 per order freight subsidy. In markets where customer acquisition cost is otherwise enormous, this subsidy is efficiently deployed. The goal is to trigger the first purchase — once a customer has bought from Target Brazil and had a positive experience, repeat purchases have meaningfully lower freight sensitivity.

HIGH · Publish Accurate Delivery Estimates Based on Actual State-Level Performance

Several states are delivering 10-20 days ahead of the estimated date shown at checkout. This gap between stated and actual performance is a missed trust-building opportunity. When a customer sees an estimated delivery of 25 days and the order arrives in 7, the positive surprise drives reviews, repeat purchases, and word-of-mouth. But when the estimate is 25 days and it arrives in 8, the customer has been pessimistically informed. Recalibrating delivery estimates to reflect actual state-level performance would improve conversion rate at checkout and increase post-purchase satisfaction.

HIGH · Expand Interest-Free Installment Messaging on High-Value Products

Brazilian consumers have a deeply ingrained preference for installment purchases. The installment data confirms this — a significant proportion of orders use multi-installment payment plans. Yet installment options are likely not prominently displayed at the product level. Adding "pay in 3x interest-free" messaging directly on product pages for items above R$200 removes a psychological barrier at the point of decision — the moment when the customer is evaluating whether to buy. Based on comparable Brazilian marketplace implementations, this change consistently lifts conversion rate on high-value products by 15-25%.

MEDIUM · Diversify Customer Base Beyond São Paulo and Southeast

SP, RJ, and MG account for approximately 60% of all customers. This concentration creates platform dependency on a single economic region. A targeted marketing investment in the northeast — particularly in Bahia, Pernambuco, and Ceará, which already appear in the top 15 customer states — would build the geographic diversification that protects Target Brazil from regional economic shocks and creates new growth vectors without competing for the same São Paulo customers.

MEDIUM · Implement Seller Quality Enforcement Based on Review Scores

The bimodal review distribution — heavy 5-star and 1-star — suggests that a small number of underperforming sellers are generating a disproportionate share of negative reviews. Identifying sellers whose order portfolio generates average scores below 3.0 and removing them from the platform protects the overall marketplace rating. A warning system for sellers below 3.5 with a 60-day improvement window creates accountability without immediately reducing seller supply.

LOW · Develop a January Recovery Campaign Targeting Gift Recipients

January is consistently the weakest revenue month across all three years — a predictable post-holiday hangover. One underutilized lever is the gift recipient segment: customers who received Target Brazil gift cards or vouchers over the holiday period but have not yet converted them. A January campaign specifically targeting voucher holders — with curated product recommendations and a time-limited incentive — would bring forward purchases that would otherwise happen in February or March, smoothing the Q1 revenue dip.

Appendix — Dataset Schema & SQL Reference

Table structures, key queries and methodology notes

Dataset Overview

Dataset	Records	Key Fields
customers.csv	99,441	customer_id, customer_unique_id, customer_city, customer_state
orders.csv	99,441	order_id, customer_id, order_status, order_purchase_timestamp, delivery dates
order_items.csv	112,650	order_id, product_id, seller_id, price, freight_value
payments.csv	103,886	order_id, payment_type, payment_installments, payment_value
products.csv	32,951	product_id, product_category_name, dimensions, weight
reviews.csv	99,224	review_id, order_id, review_score, review_creation_date
sellers.csv	3,095	seller_id, seller_city, seller_state
geolocation.csv	1,000,163	zip_code_prefix, latitude, longitude, city, state

Key SQL Queries

Revenue Growth 2017 vs 2018 (Jan–Aug):

```
SELECT
YEAR(o.order_purchase_timestamp) AS order_year,
SUM(p.payment_value) AS total_revenue,
ROUND(
(SUM(p.payment_value) - LAG(SUM(p.payment_value))
OVER (ORDER BY YEAR(o.order_purchase_timestamp)))
/ LAG(SUM(p.payment_value))
OVER (ORDER BY YEAR(o.order_purchase_timestamp)) * 100, 2
) AS pct_growth
FROM orders o
JOIN payments p ON o.order_id = p.order_id
WHERE MONTH(o.order_purchase_timestamp) BETWEEN 1 AND 8
AND YEAR(o.order_purchase_timestamp) IN (2017, 2018)
GROUP BY order_year;
```

Delivery Time and Estimate Variance per Order:

```
SELECT
order_id,
DATEDIFF(order_delivered_customer_date,
order_purchase_timestamp) AS time_to_deliver,
DATEDIFF(order_delivered_customer_date,
order_estimated_delivery_date) AS diff_estimated_delivery
FROM orders
```

```
WHERE order_delivered_customer_date IS NOT NULL
AND order_purchase_timestamp IS NOT NULL;
```

Top 5 States by Average Freight Value:

```
SELECT
c.customer_state,
ROUND(AVG(oi.freight_value), 2) AS avg_freight
FROM order_items oi
JOIN orders o ON oi.order_id = o.order_id
JOIN customers c ON o.customer_id = c.customer_id
GROUP BY c.customer_state
ORDER BY avg_freight DESC
LIMIT 5;
```

Methodology Notes

- Python (Pandas, NumPy) used for all data loading, cleaning, transformation and aggregation
- Geolocation table deduplicated to one row per zip code prefix before joining
- Delivery metrics computed only on orders with non-null delivery dates (delivered status)
- Revenue growth calculation restricted to Jan-Aug window to ensure fair year-on-year comparison
- All monetary values in Brazilian Real (R$)
- Review score average computed across all orders with at least one review response

This analysis was conducted as part of a data portfolio project demonstrating end-to-end analytical capability across SQL, Python, Excel, and business storytelling.